

Rapamycin (AY-22,989), a new antifungal antibiotic. I. Taxonomy of the producing streptomycete and isolation of the active principle.



A streptomycete was isolated from an Easter Island soil sample and found to inhibit *Candida albicans*, *Microsporum gypseum* and *Trichophyton granulosum*. The antibiotic-producing microorganism was characterized and identified as *Streptomyces hygroscopicus*. The antifungal principle was extracted with organic solvent from the mycelium, isolated in crystalline form and named rapamycin. Rapamycin is mainly active against *Candida albicans*; minimum inhibitory concentration against ten strains ranged from 0.02 to 0.2 µg/ml. Its apparent activity against *Microsporum gypseum* and *Trichophyton granulosum* is lower because of its instability in culture media on prolonged incubation required by these fungi. No activity was observed against gram-positive and gram-negative bacteria. Acute toxicity in mice is low.